Pergamon

0277-9536(95)00029-1

Soc. Sci. Med. Vol. 41, No. 11, pp. 1507–1512, 1995
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0277-9536/95 $9.50 + 0.00

SEPARATE BLACK AND WHITE INFANT MORTALITY MODELS: DIFFERENCES IN THE IMPORTANCE OF STRUCTURAL VARIABLES

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Abstract—In the United States, the disparity in black and white infant mortality persists despite reductions in overall infant mortality. In 1988, 23 of the 50 states had a black infant mortality rate that was more than twice as large as its white infant mortality rate. This study assesses whether state-level structural variables relate differentially to states' black and white infant mortality rates. With the state as the unit of analysis, separate black ($N = 34$) and white ($N = 50$) multivariate models of infant mortality were produced and compared. The structural variables accounted for 57.7% of the variance in states' black infant mortality rates and 35.2% of the variance in states' white infant mortality rates. Proportion black, percent with bachelor's degree or higher, percent below poverty, and the index of dissimilarity each made a unique contribution to the black infant mortality model. Percent with bachelor's degree or higher was the only measure that made a significant unique contribution to the white infant mortality model. Thus, although both black and white infant mortality rates were higher in states with smaller percentages of the population having a bachelor's degree or higher, black infant mortality rates were also higher in states where proportionately more black persons lived, where there were higher levels of residential segregation in the urban areas, and, contrary to what was expected, where smaller percentages of the population lived below the poverty level. This study supports the need for race-specific models of infant mortality.

Key words—infant mortality, structural variables

INTRODUCTION

In the United States, black infant mortality rates are considerably higher than white infant mortality rates. The disparity in black and white infant mortality persists despite reductions in overall infant mortality. During the period from 1940 to 1991, the United States infant mortality rate dropped from 47.0 to 8.9 infant deaths per 1000 live births, yet during that same period the ratio of black to white infant mortality rates increased from 1.7 to 2.4 [1, 2]. Similar disparities exist in state-level infant mortality rates. In 1988, the last year for which state-level race-specific data were available, white infant mortality rates ranged from 6.7 to 10.0 (all states) while black infant mortality rates ranged from 12.0 to 22.4 (among those states which had at least 20 black infant deaths) for that same year [1, p. 8]. In 1988, 23 of the 50 states had a black infant mortality rate that was more than twice as large as its white infant mortality rate [1, p. 8].

Few studies have examined state-level infant mortality [3–11]. This study assesses whether state-level structural variables relate differentially to states' black and white infant mortality rates. The structural variables are characteristics of states' social structure and political climate. In a previous study, this author found that structural variables accounted for 67.3% of the variance in state-level infant mortality rates for all races, with proportion black being the only variable that made a significant unique contribution [5]. Structural variables, however, may relate differently to state-level black vs white infant mortality. Thus, in this study, separate black and white multivariate models of infant mortality at the state level were produced and compared.

STRUCTURAL VARIABLES

Public health researchers, practitioners, and leaders have long argued that social and other structural factors must be addressed if health outcomes are to be improved [12–19]. Tesh [17] states that a structural perspective of disease prevention, i.e., one that identifies the social structure as the fundamental cause of disease, better informs public health policies than other perspectives. According to Kaplan [20], three elements of social structure are commonly examined in relation to health and illness: social identities (i.e., social positions and statuses determined by characteristics such as race and occupation), social relationships (i.e., participation in social networks and the nature of social relationships), and the broader sociocultural system (e.g., consensual values and behavior patterns). These social structure elements are believed to influence health through exposure to pathogenic or salutary circumstances, susceptibility or vulnerability to disease, and access to health care resources [20].

As already indicated, few state-level infant mortality studies have been published [3, 4, 6–11]. Infer-

ences about state-level infant mortality from studies conducted at other levels (such as the individual) can be inappropriate [21–23]. Consequently, the selection of structural variables was less grounded in prior research than desirable. Structural variables for which meaningful relationships with state-level infant mortality could be hypothesized were selected. The variables include racial distribution, educational distribution, poverty distribution, level of residential segregation, level of gender discrimination, and political climate. They are described conceptually below. With the exception of residential segregation, each structural variable's expected relationship with infant mortality is presented in terms of overall state-level infant mortality; no hypotheses regarding how the structural variables might relate differentially to black vs white infant mortality are put forth. All variables are operationalized in the following section of this paper.

States' racial, educational, and poverty distributions were expected to be related to infant mortality rates because racial minorities, the less educated, and those living in poverty tend to be in the lower social strata, and states that have proportionately more people in the lower social strata were expected to have poorer health outcomes. Living in a state in which many people are in the lower social strata may have adverse effects on health, even for those people who are not in the lower social strata themselves. Additionally, poverty describes a set of conditions which may include, for example, poor housing, poor nutrition, more crime and violence, fewer resources, and inadequate health care and, thus, describes the sociocultural system of a state's population. For these reasons, states with proportionately more racial minorities, fewer educated persons, and more persons living in poverty were expected to have higher infant mortality rates than states with proportionately smaller numbers of racial minorities, the less educated, and those living in poverty.

Level of residential segregation also describes a society's sociocultural system. As defined by Massey and Denton, "At a general level, residential segregation is the degree to which two or more groups live separately from one another, in different parts of the urban environment" [24]. A state's level of residential segregation reflects the level of residential segregation in its cities or metropolitan areas. Previous studies have found that at the city and the metropolitan area levels residential segregation is positively related to the black-white difference in infant mortality [25, 26] and is differentially related to black and white infant mortality [26–28]. Accordingly, in this study, residential segregation was expected to adversely impact black infant mortality more than white infant mortality.

Another characteristic of a sociocultural system is level of gender discrimination. Gender discrimination reflects a broader set of beliefs, values, and norms of the larger social system concerning women and is manifested in, among other things, fewer employment opportunities for women, more difficult access to resources (e.g., employment related benefits) for women, less education for women, and lower status and less power for women [29, 30]. The extent to which gender discrimination exists in a society may be related to its infant mortality because of the link between women and children (e.g., women's health during pregnancy directly influences the health of the newborn). Infant mortality was expected to be higher in states with greater gender discrimination than in states with less gender discrimination.

Finally, the political climate of a state may have implications for infant mortality in that political climate likely affects the social and health policies and programs that are supported and implemented. The political climate also reflects the larger sociocultural context of the state in that politicians and government officials are elected or appointed because of their values and beliefs. Politically liberal states are likely to be more supportive of policies and programs that provide social and health resources to their population (e.g., abortion services, school based clinics, welfare programs) than politically conservative states and, therefore, may have better health outcomes. On the other hand, politically liberal states may attract more persons in the lower social strata because of the services that are available. Even so, in this study, politically liberal states were hypothesized to have lower state-level infant mortality than politically conservative states.

METHODS

This study was cross-sectional. The data came from a variety of sources (referenced below) and in general were for the most recent year available. In some cases, measures were created from the data obtained. A more detailed description of the data and measures is provided elsewhere [3].

Racial distribution was measured by the proportion black in 1990 [31]. Educational distribution was measured by the percent of persons 25 years of age and over who (a) were high school graduates or higher in 1990 and (b) had bachelor's degree or higher in 1990 [32]. Poverty distribution was measured by the percent of all persons (except institutionalized persons, persons in military group quarters and in college dormitories, and unrelated individuals under 15 years old) with income in 1989 below the poverty level [32].

The residential segregation measure was the index of dissimilarity for 1990 (a measure of the extent to which minority group members are over and underrepresented in different areas and ranges from 0.0 to 1.0 with 1.0 being maximum residential segregation) [24, 33]. The Census Bureau calculated this index for all metropolitan areas in the United States for four groups: (1) American Indians, Eskimos, and Aleuts;

(2) Asians and Pacific Islanders; (3) Blacks; and (4) Hispanics [34]. To compute a state-level measure, the index was grouped by state (indices for the four racial categories were grouped together) and weighted and averaged [3].

Gender discrimination was measured by the ratio of male to female median earnings (for persons 16 years or more, working 35 hr or more a week for 50–52 weeks) in managerial and professional specialty occupations in 1979 [35]. Finally, political climate was measured by the number of times a Republican presidential candidate carried each state from 1968–1988 [36].

In the previous study which assessed the relationship of structural variables to state-level infant mortality for all races, three additional residential segregation measures (each measuring an additional dimension of residential segregation) were included in the analysis [5]. For this study, however, fewer independent variables could be used because only 34 states were included in the black infant mortality analyses. Consequently, the residential segregation measure most often used [24, pp. 284, 308]—the index of dissimilarity—was included in these analyses and the other three were discarded.

The dependent variable measures were (1) the black infant mortality rate averaged for 1986, 1987, and 1988, and (2) the white infant mortality rate averaged for 1986, 1987, and 1988 [1, p. 8]. The infant mortality rate is the number of infant deaths in a given year per 1000 live births in the same year. Some states have few black infant deaths and, consequently, unstable black infant mortality rates. States that had less than 20 black infant deaths in either 1986, 1987, or 1988 were excluded from the analyses for black infant mortality. These consisted of 14 states which had less than 20 black infant deaths in each year from 1986 to 1988 (Alaska, Hawaii, Idaho, Maine, Montana, New Hampshire, New Mexico, North Dakota, Oregon, Rhode Island, South Dakota, Utah, Vermont, and Wyoming), one state that had less than 20 black infant deaths in 1986 (Iowa), and one state that had less than 20 black infant deaths in 1987 (West Virginia). The remainder of the states ($N = 34$) were included in the black infant mortality analyses. All states were included in the analyses of white infant mortality ($N = 50$).

Standard multiple regressions were conducted to assess the unique contribution of each structural variable to black and white infant mortality rates separately. Plots of residuals and predicted dependent measure scores and Cook's distance statistics indicated that assumptions of normality, linearity, and homoscedasticity were met and that there were no outliers. Correlation matrices of the independent and dependent measures were produced to assist in identifying the presence of suppressor variables. If evidence existed that one or more suppressor variables were present when a significant independent variable was involved, additional analyses were performed to identify the suppressor variables. Univariate statistics were also computed.

RESULTS

Four structural measures made significant unique contributions to black infant mortality in the multivariate model: proportion black, percent with bachelor's degree or higher, percent below poverty, and the index of dissimilarity (Table 1). Except for percent below poverty, these were in the predicted direction of higher black infant mortality rates for states with larger proportions of black persons, smaller percentages of the population with bachelor's degree or higher, and higher levels of residential segregation. Contrary to what was expected, states with higher black infant mortality rates had smaller percentages of their population living in poverty. Together, the structural measures explained 57.7% of the variance in state-level black infant mortality rates (Table 1).

Black infant mortality's bivariate correlations with proportion black, percent bachelor's degree or higher, and percent below poverty (Table 2) were all substantially smaller than the standardized regression coefficients for these structural measures in the black infant mortality multivariate model (Table 1) indicating that suppressor variables were present. Subsequent analyses suggested that (1) percent high

Table 1. Standard multiple regression models of black ($N = 34$) and white ($N = 50$) infant mortality rates, averaged for 1986–1988

	Black infant mortality rates			White infant mortality rates		
Independent variable	Unstandardized regression coefficient	Standardized regression coefficient	sr^2 (unique)	Unstandardized regression coefficient	Standardized regression coefficient	sr^2 (unique)
Proportion black	12.45*	0.59*	0.152*	2.36	0.35	0.055
% High school or higher	0.14	0.41	0.034	0.02	0.22	0.010
% Bachelor's degree or higher	−0.26†	−0.57†	0.125†	−0.07†	−0.42†	0.073†
% Below poverty	−0.30*	−0.67*	0.148*	0.03	0.19	0.015
Index of dissimilarity	6.95†	0.38†	0.106†	−0.90	−0.23	0.027
Ratio of male to female earnings	0.39	0.01	0.000	−0.40	−0.04	0.001
# Times Republican presidential candidate carried state	0.24	0.16	0.018	0.08	0.18	0.022
Intercept	8.96			8.46		
R^2			0.577*			0.352*

*$P < 0.01$.
†$P < 0.05$.

Table 2. Correlation matrices, means, and standard deviations for structural variables and black ($N = 34$) and white ($N = 50$) infant mortality rates, average for 1986–1988*

Variables*	BLACK	HS+	BA+	POV	DISS	M:F	REP	WIMR	BIMR
Proportion (BLACK)		−0.66†	−0.21	0.39†	0.51†	0.26	−0.22	0.20	
% High school or higher (HS+)	−0.70†		0.63†	−0.70†	−0.51†	−0.15	0.28‡	−0.24	
% Bachelor's degree or higher (BA+)	−0.27	0.68†		−0.66†	−0.22	0.02	−0.00	−0.44†	
% Below poverty (POV)	0.51†	−0.77†	−0.65†		0.17	0.06	−0.05	0.40†	
Index of dissimilarity (DISS)	0.25	−0.27	−0.16	0.02		0.28‡	−0.41†	−0.12	
Ratio of male to female earnings (M:F)	0.06	−0.01	0.10	−0.03	0.05		0.23	−0.00	
# Times Republican presidential candidate carried state (REP)	−0.19	0.07	−0.11	−0.08	−0.27	0.33		0.25	
Average black infant mortality rate (BIMR)	0.18	0.04	−0.09	−0.32	0.45†	0.08	0.09		
Average white infant mortality rate (WIMR)									
Analysis with 34 states									
Mean	0.13	75.0	19.7	13.3	0.55	1.7	4.7		17.5
Standard deviation	0.09	5.5	4.0	4.3	0.10	0.06	1.3		1.9
Analysis with 50 states									
Mean	0.10	76.3	19.8	13.1	0.48	1.7	4.8	8.8	
Standard deviation	0.09	5.6	3.7	4.1	0.16	0.07	1.4	0.6	

*Lower left-hand portion of matrix presents correlations using data for 34 states; upper right-hand portion of matrix presents correlations using data for 50 states.
†$P < 0.01$.
‡$P < 0.05$.

school graduate or higher and percent below poverty suppressed irrelevant variance and enhanced the importance of both proportion black and percent with bachelor's degree or higher in explaining black infant mortality; and (2) proportion black, percent high school graduate or higher, and percent bachelor's degree or higher suppressed irrelevant variance and enhanced the importance of percent below poverty in explaining black infant mortality.

Percent with bachelor's degree or higher was the only measure that made a significant unique contribution to white infant mortality in the multivariate model (Table 1). Its relationship was in the predicted direction of higher white infant mortality rates for states with a smaller percentage of their population having a bachelor's degree or higher. Altogether, the structural measures explained 35.2% of the variance in state-level white infant mortality rates (Table 1).

Although percent below poverty had a significant positive bivariate relationship with white infant mortality (Table 2), it did not make a significant unique contribution to white infant mortality in the multivariate model. Thus, its relationship with white infant mortality was accounted for by other measures in the multivariate model.

DISCUSSION

The structural measures included in this study accounted for substantially more variance in black infant mortality rates (57.7%) than in white infant mortality rates (35.2%) at the state level. Although percent with bachelor's degree or higher was significant in both the black and white multivariate models, three additional variables were significant in the black model. Both black and white infant mortality rates were higher in states with smaller percentages of the population having a bachelor's degree or higher, but black infant mortality rates were also higher in states where proportionately more black persons lived, where there were higher levels of residential segregation in the urban areas, and, contrary to what was expected, where smaller percentages of the population lived below the poverty level.

Proportion black's significance in the black infant mortality model is consistent with the earlier study that examined the relationship of structural variables to state-level infant mortality for all races (all states) [5]. It was hypothesized that proportion black would be related to state-level infant mortality because racial minority groups tend to be in the lower social strata, and states with relatively more persons in the lower social strata were expected to have higher infant mortality rates. The other structural variables included in this study did not completely account for the relationship between proportion black and black infant mortality—states with relatively more black persons have higher black infant mortality rates for some reason other than that these states also have proportionately more people living in poverty and fewer with high levels of education. Future studies, however, should examine whether the other structural variables more strongly mediate the proportion black and black infant mortality relationship when race-specific measures (e.g., percent of black persons living in poverty) are used.

In a previous study, health services (e.g., availability of abortion providers and family practitioners) partially accounted for the relationship between structural variables and infant mortality rates for all races at the state level [3, 4]. Health services could also account for the proportion black and black infant mortality relationship. For example, it may be that states with larger proportions of black persons attract fewer family practitioners, and, in turn, having fewer family practitioners state-wide may adversely impact state-level black infant mortality. Future research should also identify other variables that might account for the proportion black and black infant mortality relationship. For example, it

may be that the states with relatively more black persons are also the states with worse environmental conditions (e.g., hazardous waste sites) or with relatively more black smokers, and these factors may influence state-level black infant mortality.

This study's findings add to the existing evidence that residential segregation adversely affects black infant health [25–28]. Residential segregation may influence state-level black infant mortality through, for example, limited access to health services or concentration of adverse physical or environmental conditions (e.g., poor housing, air pollution) in highly segregated areas. The experience of residential segregation itself may adversely impact health. Additional studies that examine the direct and indirect health effects of residential segregation are needed. Policies that might reduce the level of residential segregation should be explored.

The negative relationship between poverty and black infant mortality was not expected and requires further examination. It could be that states with smaller percentages of the total population living in poverty are states with smaller percentages of whites living in poverty. That is, states with smaller percentages living in poverty may be states in which poverty is concentrated within the black population. As indicated above, it would be helpful to assess the relationship between race-specific poverty and race-specific infant mortality.

This paper focuses on the differences in separate state-level black and white infant mortality multivariate models. The significance of percent with bachelor's degree or higher in both the black and white models, however, should not be overlooked. The extent to which a state's population is college-educated was related to both its black and its white infant mortality rates suggesting that by increasing educational opportunities beyond high school for all people, regardless of race, states may improve both black and white infant mortality. This, however, was not the case for high school education. Furthermore, the ratio of male to female earnings and the number of times Republican presidential candidates carried the state were not related to either black or white infant mortality rates. However, because measures of state-level gender discrimination and political climate were limited, identifying other measures of these variables may be worthwhile for future studies. Additionally, gender discrimination differs for black and white women [37] and, thus, race-specific gender discrimination measures may be more strongly related to race-specific infant mortality than the overall gender discrimination measure used here.

The two multiple regressions presented differed in that data for 34 states were analyzed in the black infant mortality regression whereas data for all 50 states were analyzed in the white infant mortality regression. When only the subset of 34 states were included in analysis of white infant mortality, about the same amount of variance was explained (33.1%, compared to 35.2% when all 50 states are included), but none of the individual structural variables were significant (data not shown). Though the dependent variable measure used in this study did not permit analyzing black infant mortality in all 50 states, future studies could do so by using, for example, a measure that pooled black infant deaths for multiple years. Findings from analyses that include all 50 states may differ from those presented here.

A large percentage of the variance in both the black and white infant mortality rates was unexplained. A difficulty in doing state-level analyses is that the number of independent variables that can be included in any one multiple regression is limited. Future research could exclude the less important structural measures (e.g., percent high school graduate or higher, ratio of male to female earnings) and include other state-level variables such as measures of unemployment, racial discrimination, marital status, and teenage pregnancy. Additionally, potential confounding variables (e.g., percent living in urban areas) of the structural and black and white infant mortality relationships could be examined.

For this study, structural variables that characterized the population of each state were examined in relation to state-level black and white infant mortality rates. The findings indicate that structural variables do differentially relate to states' black and white infant mortality rates. Black infant mortality rates were higher in states with relatively more black persons, states with higher levels of residential segregation, and relatively fewer persons living in poverty. These structural factors were not significantly related to white infant mortality when considered in a multivariate model. This study supports the need for race-specific models of infant mortality. Understanding the differences in the importance of structural variables for black and white infant mortality is necessary if reductions in infant mortality, particularly black infant mortality, are to be achieved.

Acknowledgements—Thanks to Susan T. Ennett and Lee Ann Kaskutas who provided helpful comments on an earlier draft of this paper.

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